

performance. This is especially important for multi-band radar systems and 5G networks operating in millimetre-wave bands.

#### **Compact and lightweight designs.**

The increasing demand for smaller, lighter systems that reduce payload costs in satellites and enable portable radar and 5G equipment adds another layer of complexity. Engineers need solutions that integrate high-performance components into compact designs.

### **The rise of GaN-on-SiC MMIC power amplifiers**

GaN-on-SiC MMIC power amplifiers have emerged as a groundbreaking technology in addressing these challenges. They combine the unique material properties of GaN and SiC to deliver unprecedented performance and efficiency.

#### **What makes GaN-on-SiC special?**

GaN-on-SiC amplifiers stand out due to their superior material properties and enabling unmatched power efficiency. Here is what sets them apart:

**Wide bandgap and high power density.** GaN's wide bandgap allows for higher breakdown voltage, enabling devices to handle more power in smaller footprints.

**Exceptional thermal conductivity.** SiC substrates provide excellent thermal dissipation, reducing the risk of overheating and ensuring reliable operation under high-power conditions.

**Broadband capabilities.** GaN-based devices can operate across wide frequency bands, making them ideal for multi-band applications in radar and communication systems.

**Monolithic integration.** MMIC technology enables the integration of multiple components into a single chip, reducing size, weight, and complexity.

### **Applications in military satellites and radar systems**

Military satellite communication and radar technologies have stringent

power, reliability, and precision requirements. GaN-on-SiC MMIC power amplifiers excel in these areas, revolutionising how these systems operate.

#### **Long-distance communication.**

GaN-on-SiC amplifiers can transmit signals over long distances with minimal power loss, a critical feature for military satellites that must maintain global communication. The high efficiency of these amplifiers ensures that power usage is optimised without compromising signal strength.

#### **Enhanced radar capabilities.**

Radars rely on high-power amplifiers for superior range and resolution. GaN-on-SiC devices enable radars to detect smaller objects at greater distances, improving accuracy in tracking and identification. Additionally, their broadband capabilities allow for better adaptation to varying operational frequencies.

**Rugged performance in harsh environments.** Military applications often demand equipment capable of functioning in extreme temperatures and conditions. The robustness of GaN-on-SiC amplifiers makes them ideal for deployment in such environments, ensuring consistent performance under stress.

### **Impact on 5G network communication**

5G networks are revolutionising telecommunications, but they bring significant technical challenges, especially regarding power efficiency, latency, and frequency operation. GaN-on-SiC MMIC amplifiers provide a practical solution to these challenges.

#### **Supporting ultra-high speeds.**

The clarity and power efficiency of GaN-on-SiC amplifiers enhances signal transmission and reception, which are vital for achieving the ultra-high-speed data rates promised by 5G networks.

**Reducing latency.** By minimising signal distortion and power losses, these

amplifiers contribute to lower latency, ensuring smoother communication for critical applications like autonomous vehicles and remote surgery.

**Enabling millimetre-wave operation.** 5G networks often operate in the millimetre-wave spectrum, which requires amplifiers that can handle high frequencies. GaN-on-SiC amplifiers meet this requirement while maintaining power efficiency.

**Sustainable infrastructure.** The energy efficiency of GaN-on-SiC amplifiers reduces power consumption, lowers operational costs, and supports the creation of sustainable and environmentally friendly network infrastructure.

### **Challenges and prospects**

Despite their advantages, GaN-on-SiC MMIC power amplifiers face challenges, such as high manufacturing costs and the need for further advancements in materials and fabrication processes. Scaling production while maintaining quality will be critical for broader adoption.

Researchers are actively working to improve device reliability and reduce costs. Innovations such as advanced packaging techniques and hybrid materials could further enhance the performance of GaN-on-SiC devices, making them more accessible across industries.

GaN-on-SiC MMIC power amplifiers represent a transformative technology for military satellite communication, radar systems, and 5G networks. Their ability to deliver high power, operate across wide bandwidths, and manage thermal challenges positions them as a cornerstone of next-generation communication systems. As development continues, this technology is set to redefine benchmarks for efficiency, performance, and reliability in critical communication applications, bridging the gap between today's challenges and tomorrow's solutions. **EFY**